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```
MariaDB [(none)]> create database hotel_info;  
Query OK, 1 row affected (0.005 sec)
```

```
MariaDB [(none)]> use hotel_info;  
Database changed
```

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```
MariaDB [hotel_info]> CREATE TABLE Hotel ( hotelNo VARCHAR(10) PRIMARY KEY,  
hotelName VARCHAR(50) NOT NULL, city VARCHAR(50) NOT NULL );  
Query OK, 0 rows affected (0.005 sec)
```

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====

```
MariaDB [hotel_info]> CREATE TABLE Guest ( guestNo VARCHAR(10) PRIMARY KEY,  
guestName VARCHAR(50) NOT NULL, guestAddress VARCHAR(100) NOT NULL );  
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [hotel_info]> CREATE TABLE Room (  
->     roomNo INT,  
->     hotelNo VARCHAR(10),  
->     type VARCHAR(10),  
->     price DECIMAL(5,2),  
->     PRIMARY KEY (roomNo, hotelNo),  
->     FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),  
->     CONSTRAINT chk_type CHECK (type IN ('Single', 'Double', 'Family')),  
->     CONSTRAINT chk_price CHECK (price BETWEEN 10 AND 100),  
->     CONSTRAINT chk_roomNo CHECK (roomNo BETWEEN 1 AND 100)  
-> );  
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [hotel_info]> CREATE TABLE Booking (  
->     hotelNo VARCHAR(10),  
->     guestNo VARCHAR(10),  
->     dateFrom DATE,  
->     dateTo DATE,  
->     roomNo INT,  
->     PRIMARY KEY (hotelNo, guestNo, dateFrom),
```

```

-> FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),
-> FOREIGN KEY (guestNo) REFERENCES Guest(guestNo),
-> FOREIGN KEY (roomNo, hotelNo) REFERENCES Room(roomNo, hotelNo),
-> CONSTRAINT chk_dates CHECK (dateTo > dateFrom)
-> );

```

Query OK, 0 rows affected (0.014 sec)

MariaDB [hotel_info]> DELIMITER //

MariaDB [hotel_info]>

MariaDB [hotel_info]> CREATE TRIGGER PreventPastBookings

```

-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF NEW.dateFrom <= CURRENT_DATE OR NEW.dateTo <= CURRENT_DATE THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Error: Booking dates must be in the future.';
->     END IF;
-> END; //

```

Query OK, 0 rows affected (0.019 sec)

MariaDB [hotel_info]>

MariaDB [hotel_info]>

MariaDB [hotel_info]> CREATE TRIGGER PreventRoomDoubleBooking

```

-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF EXISTS (
->         SELECT 1 FROM Booking
->         WHERE roomNo = NEW.roomNo
->             AND hotelNo = NEW.hotelNo
->             AND dateFrom < NEW.dateTo
->             AND dateTo > NEW.dateFrom
->     ) THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Error: Room is already double-booked for these
dates.';
->     END IF;
-> END; //

```

Query OK, 0 rows affected (0.016 sec)

MariaDB [hotel_info]>

MariaDB [hotel_info]>

MariaDB [hotel_info]> CREATE TRIGGER PreventGuestOverlap

```

-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF EXISTS (

```

```

->         SELECT 1 FROM Booking
->         WHERE guestNo = NEW.guestNo
->             AND dateFrom < NEW.dateTo
->             AND dateTo > NEW.dateFrom
->     ) THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Error: Guest already has a booking during these
dates.';
->     END IF;
-> END; //
```

Query OK, 0 rows affected (0.011 sec)

```

MariaDB [hotel_info]> DELIMITER ;
MariaDB [hotel_info]>
```

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====

```

MariaDB [hotel_info]> CREATE TABLE BookingArchive (
->     hotelNo VARCHAR(10),
->     guestNo VARCHAR(10),
->     dateFrom DATE,
->     dateTo DATE,
->     roomNo INT
-> );
```

Query OK, 0 rows affected (0.008 sec)

-- am doing this so that the views will have some data instead of empty things

```

MariaDB [hotel_info]> INSERT INTO Hotel VALUES
-> ('H01', 'Kigali Serena', 'Kigali'),
-> ('H02', 'Accra City Hotel', 'Accra'),
-> ('H03', 'London Grand', 'London');
```

Query OK, 3 rows affected (0.005 sec)

Records: 3 Duplicates: 0 Warnings: 0

```

MariaDB [hotel_info]> INSERT INTO Guest VALUES
-> ('G01', 'Alice Smith', '123 Main St, London, UK'),
-> ('G02', 'Carlos Silva', '456 Paulista Ave, Sao Paulo, Brazil'),
-> ('G03', 'Kwame Mensah', '789 Independence Ave, Accra, Ghana');
```

Query OK, 3 rows affected (0.005 sec)

Records: 3 Duplicates: 0 Warnings: 0

```

MariaDB [hotel_info]> INSERT INTO Room VALUES
-> (10, 'H01', 'Single', 50.00),
```

```
-> (11, 'H01', 'Double', 80.00),
-> (20, 'H02', 'Double', 90.00),
-> (30, 'H03', 'Family', 100.00);
Query OK, 4 rows affected (0.007 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
-- here i runned into error where I
-- was not able to add the past booking dates because of the triggers
-- is disabled it using drop trigger PreventPastBookings;
-- so that i can be able to test the archiving scripts
```

```
MariaDB [hotel_info]> INSERT INTO Booking VALUES
-> ('H01', 'G01', '2012-05-10', '2012-05-15', 10),
-> ('H02', 'G03', '2026-12-01', '2026-12-05', 20),
-> ('H03', 'G02', '2027-01-10', '2027-01-20', 30);
Query OK, 3 rows affected (0.012 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
-- Now I am putting back on the triggers for booking table
```

```
MariaDB [hotel_info]> DELIMITER //
MariaDB [hotel_info]>
MariaDB [hotel_info]> CREATE TRIGGER PreventPastBookings
-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF NEW.dateFrom <= CURRENT_DATE OR NEW.dateTo <= CURRENT_DATE THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Error: Booking dates must be in the future.';
->     END IF;
-> END; //
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [hotel_info]>
MariaDB [hotel_info]> DELIMITER ;
```

```
MariaDB [hotel_info]> INSERT INTO BookingArchive
-> SELECT * FROM Booking WHERE dateFrom < '2013-01-01';
Query OK, 1 row affected (0.005 sec)
Records: 1 Duplicates: 0 Warnings: 0
```

```
MariaDB [hotel_info]>
MariaDB [hotel_info]> DELETE FROM Booking WHERE dateFrom < '2013-01-01';
Query OK, 1 row affected (0.005 sec)
```

```
-- Updatable View 1
```

```
MariaDB [hotel_info]> CREATE VIEW KigaliHotels AS
-> SELECT * FROM Hotel WHERE city = 'Kigali';
Query OK, 0 rows affected (0.007 sec)
```

```
MariaDB [hotel_info]> select * from kigalihotels;
+-----+-----+-----+
| hotelNo | hotelName      | city   |
+-----+-----+-----+
| H01     | Kigali Serena | Kigali |
+-----+-----+-----+
1 row in set (0.010 sec)
```

-- Updatable view 2

```
MariaDB [hotel_info]> CREATE VIEW SingleRooms AS
-> SELECT * FROM Room WHERE type = 'Single';
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [hotel_info]> select * from singlerooms;
+-----+-----+-----+-----+
| roomNo | hotelNo | type   | price |
+-----+-----+-----+-----+
| 10     | H01     | Single | 50.00 |
+-----+-----+-----+-----+
1 row in set (0.002 sec)
```

-- non Updatable view 1

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```
MariaDB [hotel_info]> CREATE VIEW CheapestHotels AS
-> SELECT h.hotelNo, h.hotelName, h.city, r.price
-> FROM Hotel h
-> JOIN Room r ON h.hotelNo = r.hotelNo
-> WHERE r.price = (SELECT MIN(price) FROM Room);
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [hotel_info]> select * from cheapesthotels;
+-----+-----+-----+-----+
| hotelNo | hotelName      | city   | price |
+-----+-----+-----+-----+
| H01     | Kigali Serena | Kigali | 50.00 |
+-----+-----+-----+-----+
1 row in set (0.006 sec)
```

-- non Updatable view 2

```
MariaDB [hotel_info]> CREATE VIEW BookingSummary AS
-> SELECT b.dateFrom, b.dateTo, g.guestName, h.hotelName
-> FROM Booking b
-> JOIN Guest g ON b.guestNo = g.guestNo
-> JOIN Hotel h ON b.hotelNo = h.hotelNo;
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [hotel_info]> select * from bookingsummary;
+-----+-----+-----+-----+
| dateFrom | dateTo | guestName | hotelName |
+-----+-----+-----+-----+
| 2026-12-01 | 2026-12-05 | Kwame Mensah | Accra City Hotel |
| 2027-01-10 | 2027-01-20 | Carlos Silva | London Grand |
+-----+-----+-----+-----+
2 rows in set (0.002 sec)
```

7.14
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```
MariaDB [hotel_info]> CREATE VIEW BRICSGuests AS
-> SELECT * FROM Guest
-> WHERE guestAddress LIKE '%Brazil%'
-> OR guestAddress LIKE '%Russia%'
-> OR guestAddress LIKE '%India%'
-> OR guestAddress LIKE '%China%'
-> OR guestAddress LIKE '%South Africa%';
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [hotel_info]> select * from bricsguests;
+-----+-----+-----+
| guestNo | guestName | guestAddress |
+-----+-----+-----+
| G02 | Carlos Silva | 456 Paulista Ave, Sao Paulo, Brazil |
+-----+-----+-----+
1 row in set (0.003 sec)
```

7.15
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-- create users

```
MariaDB [hotel_info]> CREATE USER 'Manager'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.018 sec)
```

```
MariaDB [hotel_info]> CREATE USER 'Director'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.001 sec)
```

```
MariaDB [hotel_info]> CREATE USER 'Accounts'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.005 sec)
```

```
-- Grant them privileges
```

```
MariaDB [hotel_info]> GRANT ALL PRIVILEGES ON CheapestHotels TO
'Manager'@'localhost', 'Director'@'localhost' WITH GRANT OPTION;
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [hotel_info]> GRANT ALL PRIVILEGES ON BRICSGuests TO 'Manager'@'localhost',
'Director'@'localhost' WITH GRANT OPTION;
Query OK, 0 rows affected (0.002 sec)
```

```
7.16
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```

```
-- Grant and then revoke Accounts privileges
```

```
MariaDB [hotel_info]> GRANT SELECT ON CheapestHotels TO 'Accounts'@'localhost';
Query OK, 0 rows affected (0.006 sec)
```

```
MariaDB [hotel_info]> GRANT SELECT ON BRICSGuests TO 'Accounts'@'localhost';
Query OK, 0 rows affected (0.001 sec)
```

```
MariaDB [hotel_info]>
MariaDB [hotel_info]> REVOKE SELECT ON CheapestHotels FROM 'Accounts'@'localhost';
Query OK, 0 rows affected (0.001 sec)
```

```
MariaDB [hotel_info]> REVOKE SELECT ON BRICSGuests FROM 'Accounts'@'localhost';
Query OK, 0 rows affected (0.002 sec)
```